



How yoga affects your brain

This chart looks at the neuroscience that may explain the vast mental and physical benefits of yoga. Modern science shows us that the brain maintains its ability to adapt across a lifetime, making it possible to break bad habits and negative patterns. It can also create the key chemicals that pharmaceutical companies synthesize in a lab. Research is uncovering the huge potential of yoga therapy to help people on a global scale. These effects stem from yoga's multidimensional approach, reflected in its 8-limb structure (see p.198), which includes guidelines on self-control and self-regulation.

↑ Brain alpha wave activity increased

Alpha waves are associated with relaxation.

↑ **GABA increased** Gamma-aminobutyric acid counteracts anxiety and stress symptoms, leading to more relaxation.

↑ **Serotonin increased** Serotonin helps regulate your mood. Low levels of usable serotonin are associated with depression.

↑ **BDNF increased** Brain-derived neurotrophic factor is a protein responsible for neuron health and neuroplasticity. Yoga can boost levels of BDNF, which may help people with chronic pain or depression.

↻ Dopamine regulated

Dopamine acts as your body's reward system and dysfunction is associated with addiction. Research suggests that meditation results in improved self-regulation.

↓ **Cortisol reduced** Cortisol is a stress hormone. When your baseline increases and levels are too high for too long, it can lead to inflammation and weight gain.

↓ **Norepinephrine reduced** A decrease in norepinephrine, or adrenaline, means fewer stress hormones in your system.

